Fulton Science Math Tournament

October 2026 - Relay Round 1

Honor Pledge

As a participant of FSMT, I promise to act with honesty, integrity, and respect in every challenge and competition I face. I will not seek or offer unauthorized assistance, and I will ensure that all work I submit is truly my own. I understand that fairness and trust are essential to the spirit of mathematics, and I accept full responsibility for upholding these values. I acknowledge that any breach of this pledge may lead to disqualification, and I am committed to competing with honor, diligence, and respect for my fellow competitors.

Team Captain's Name _____ Date _____

School _____

Team Name _____

PLEASE DO NOT BEGIN UNTIL BEING INSTRUCTED TO DO SO

The same four-student team will compete in a Relay Round consisting of four connected problems. After the first question, each following question requires the answer(s) from the previous one(s) to solve. All team members will have access to the full set of problems, and the team will submit answers to all four. Points are awarded based on consecutive correct answers: 10 points for the first correct, 15 for the first two, 25 for the first three, and 40 points for answering all four correctly. This round has a 10-minute time limit.

It is important to note that in every written round of the competition, the answer must include the unit specified in the answer blank. These units are always presented in their plural form, regardless of whether a singular form might seem more appropriate based on the context of the problem. Answers that fail to include the exact unit as indicated will not be accepted. For additional clarification and a comprehensive explanation of rules, procedures, and scoring, participants should consult the Rules and Format Page provided. If you are still uncertain about a question, an answer, or the required format, ask a proctor for clarification immediately.

Team Score _____ **Scorer's Initials** _____

1-1. _____ integers A number is called coprime to another number if the only positive divisor they share is 1 (i.e., their greatest common factor is 1). How many positive integers less than 20 are coprime to 20?

1-2. _____ units^2 Let T be the answer to the previous problem. An isosceles trapezoid $ABCD$ has parallel bases $\overline{AB}$ and $\overline{CD}$, with $AB = T$, $CD = 2T$, and legs $BC = AD = 5$. What is the area of the trapezoid?

1-3. _____ Let T be the answer to the previous problem. Jeffery calculates the sum of the first $\sqrt{T}$ positive perfect squares, while Robert calculates the sum of the first T positive integers. What is the absolute difference between their sums?

1-4. _____ Let T be the answer to the previous problem. Let N be the smallest positive integer greater than T that is divisible by the sum of the digits of T. Compute the sum of the distinct prime factors of N.

Fulton Science Math Tournament

October 2026 - Relay Round 2

Honor Pledge

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Team Captain's Name _____ Date _____

School _____

Team Name _____

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Team Score _____ **Scorer's Initials** _____

2-1. _____ We introduce a new operation, $\diamondsuit$, where $(a \diamondsuit b) = 2ab + (a+b)(a-b)$. For example, $(2 \diamondsuit 1) = 2(2)(1) + (2+1)(2-1) = 7$. What is the value of $(7 \diamondsuit 3)$?

2-2. _____ Let T be the answer to the previous problem. Two positive integers have a sum of T. The smaller integer is 4 more than half the larger integer. Find the larger integer.

2-3. _____ meters Let T be the answer to the previous problem. Martin wants to build a square fence around his circular yard. The square must completely contain the circle and have the smallest possible area. He knows that the area of the circle is $\frac{T}{13}\pi$ **km**^{2}. What side length should he make the square in **meters**? (Note: 1 kilometer is equal to 1000 meters)

2-4. _____ Let T be the answer to the previous problem, and let Q be the sum of the digits of T. A student has Q distinct math books and 3 distinct science books. The student wants to arrange all $Q+3$ books in a row on a shelf so that no science books are directly next to each other. How many different arrangements are possible?